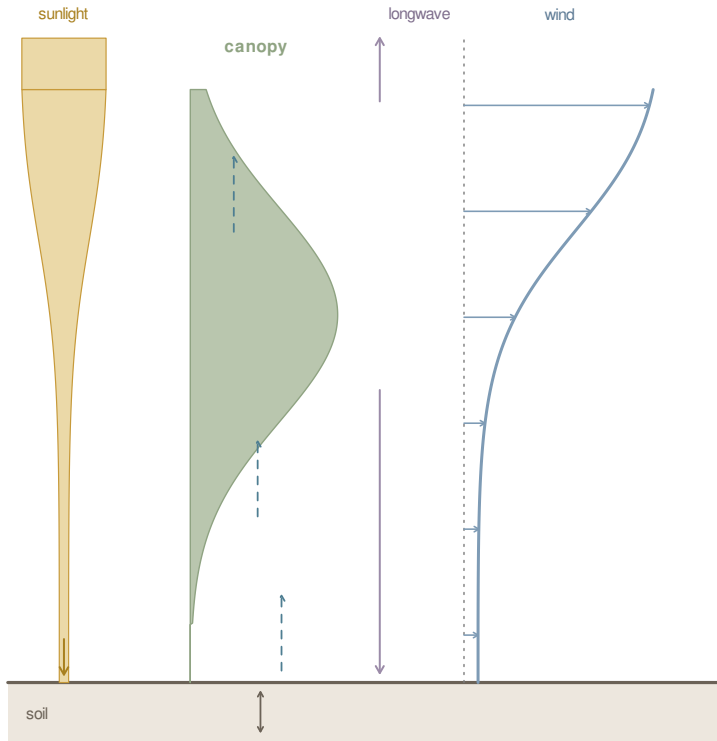




- ① Shortwave interception. Radiation is attenuated with cumulative leaf area, so less energy reaches the floor as the canopy thickens.
- ② Longwave exchange. Foliage emits downward and screens the understory from a cold sky, so it warms the understory at night.
- ③ Turbulent transfer. Drag decays the wind with depth, insulating the sub-canopy air but also trapping the heat released below it.
- ④ Latent heat. Energy consumed by evaporation, from crown, understory and soil, does not warm the air, and raises its humidity instead.
- ⑤ Soil heat storage. The ground absorbs by day and releases at night, so soil and air offsets are correlated but not interchangeable.

Every term acts at the height where the foliage sits, so a canopy is three-dimensional, not a single number.